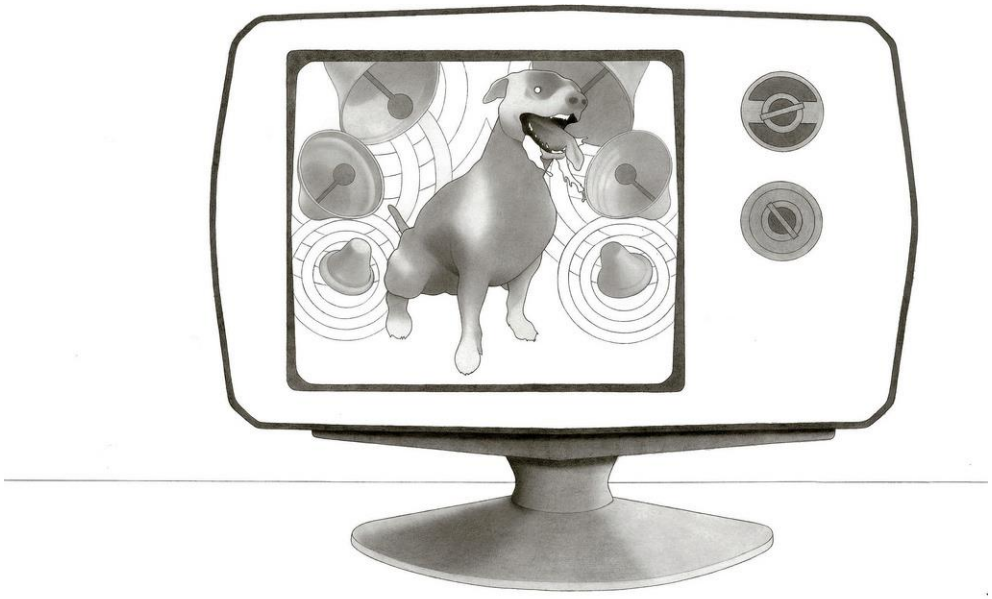
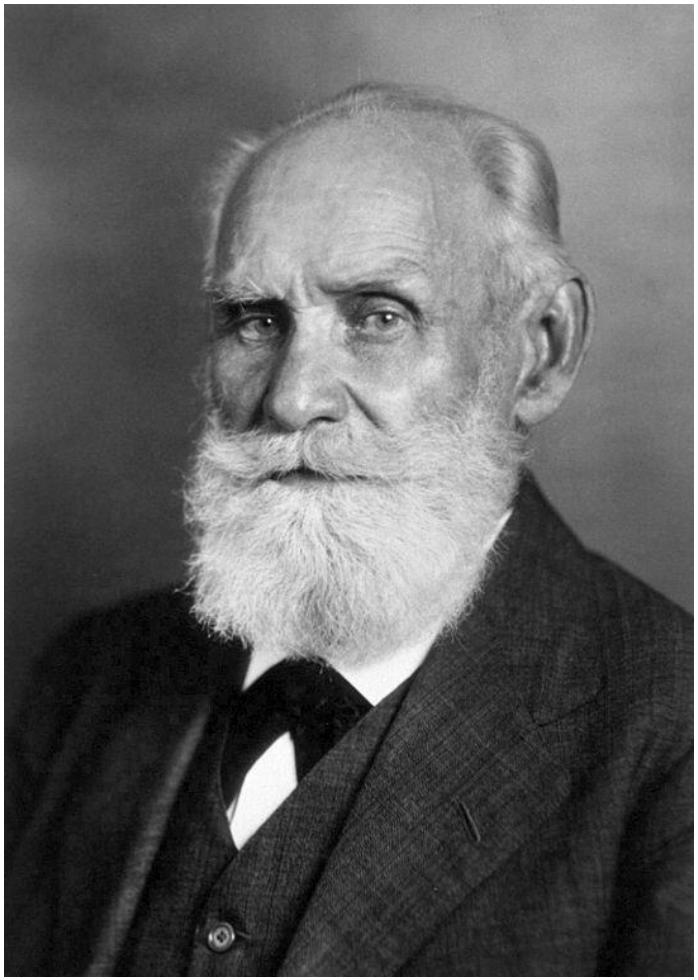


Learning & Behavior

Yashpal Jogdand
[AHUL261]



Behaviorism



Pavlov



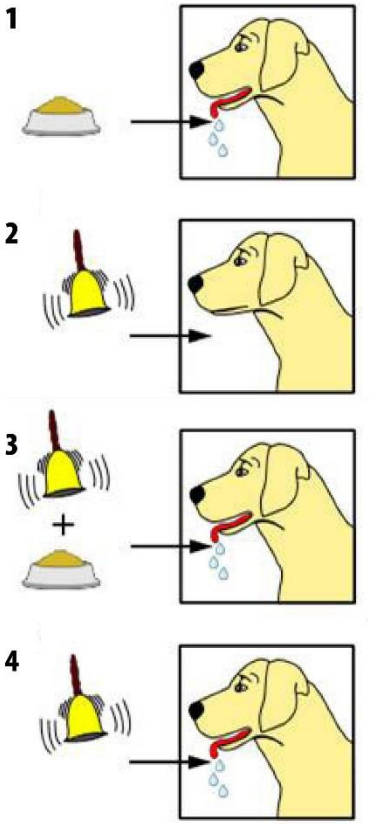
Skinner

Pavlov's Experiment

- Originally
- CS (Bell) → Nothing
- US (Meat) → UR (Salivating)

- Many Trials
- CS (Bell) : US (Meat) → UR (Salivating)

- Eventually
- CS (Bell) → CR (Salivating)



Classical Conditioning Components

Unconditioned Stimulus (US) – A stimulus that elicits a particular response without prior training

Unconditioned Response (UR) – A response that occurs to a stimulus without prior training

Conditioned Stimulus (CS) – A stimulus that does not elicit a particular response initially, but comes to do so as a result of CC

Conditioned Response (CR) – The response that is eventually elicited by the conditioned stimulus after repeated pairings with the unconditioned stimulus



Classical Conditioning Examples



- "The Office" Pavlovian Experiment

- **Discussion: What examples of CC can you think of from your own life?**



- Take out a blank sheet of paper
- Take out a pen
- All books and binders on the floor
- Put cell phones away
- Pens ready…
- RELAX!

Conditioned Emotional Response



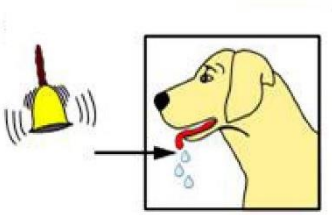
- Originally
- CS (Instructions) → Nothing
- US (Quiz) → UR (Anxiety)

- Many trials
- CS (Instructions) : US (Quiz) → UR (Anxiety)

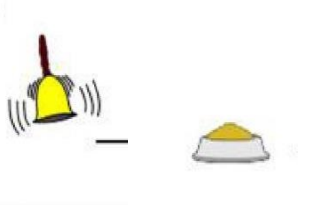
- Eventually
- CS (Instructions) → CR (Anxiety)

Extinction

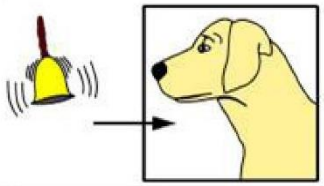
- After conditioning
- CS (Bell) → CR (Salivating)



- Many Trials
- CS (Bell) : US removed (No meat)

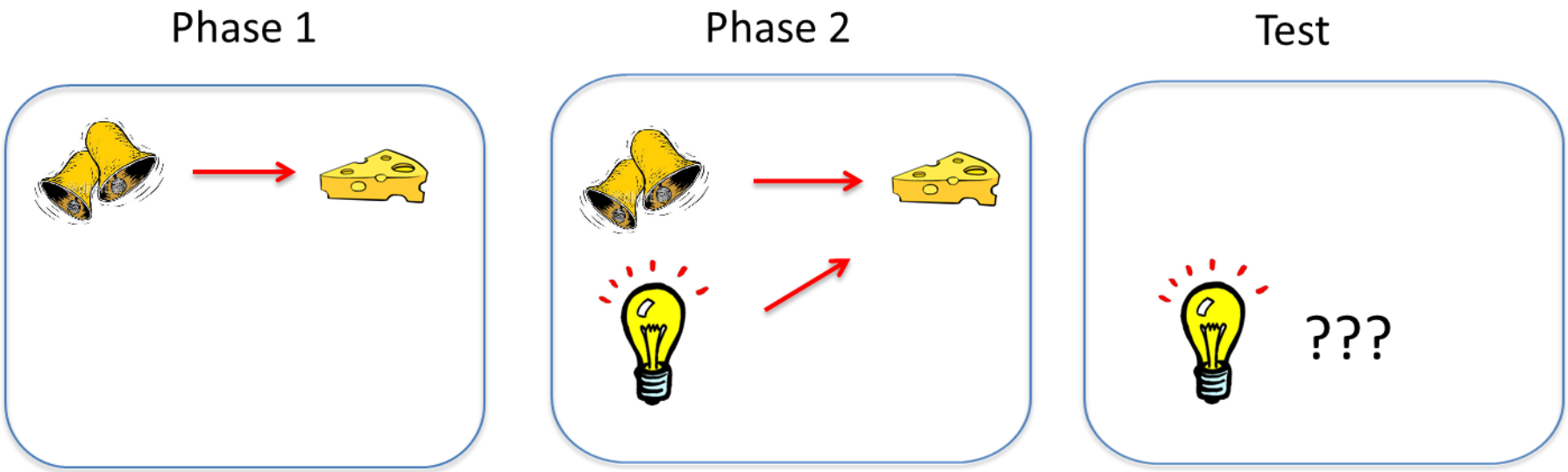


- Eventually
- CS (Bell) → Nothing (extinction)



Blocking

Blocking – Occurs when a previous association prevents another association from being formed.



Example of Blocking

The neutral manager must give an unpopular announcement

- US = Pay Cut Announcement
- UR = Displeasure of staff with announcement
- CS1 = Steve
- CS2 = Jim
- CR = Displeasure at sight of Steve

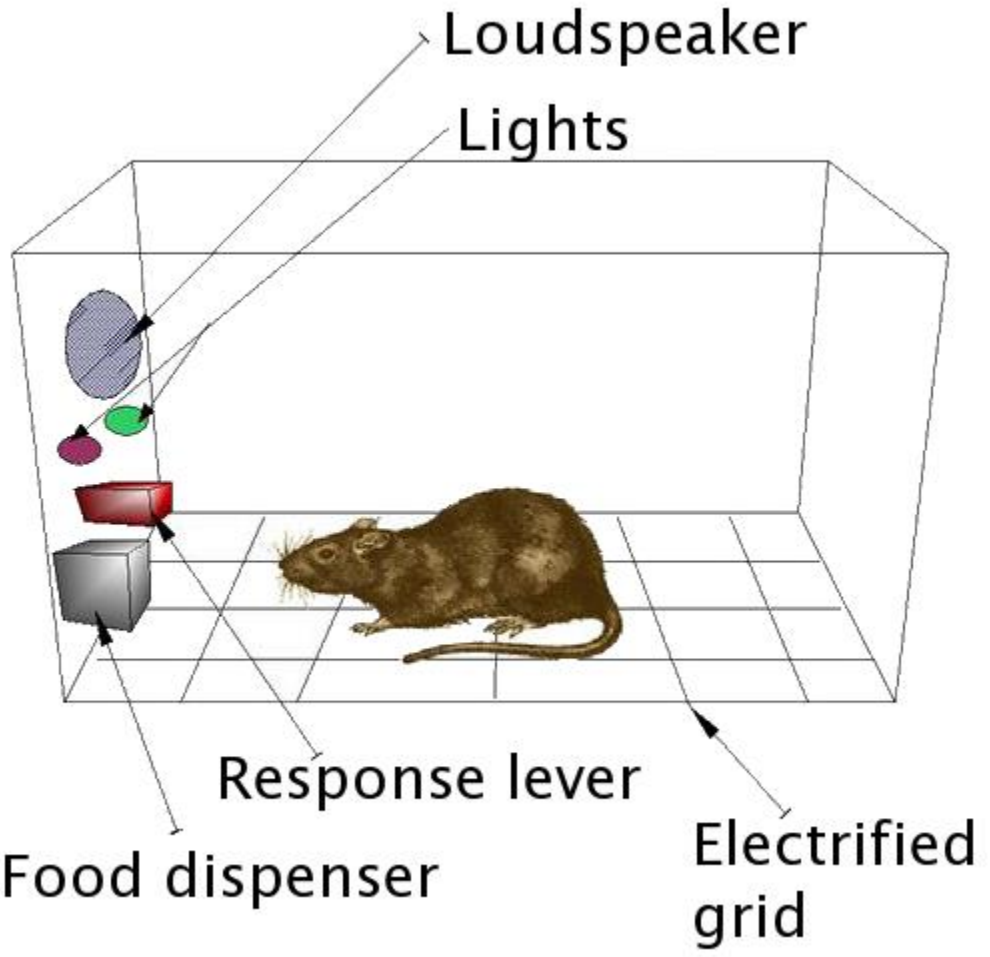
What could Jim do?

4 Things to Know about CC

1. CC generates multiple responses.
2. CC establishes preferences AND aversions.
3. CC underlies various psychological conditions.
4. Pairing the US and CS is NOT sufficient for conditioning to occur.

Operant Conditioning

Operant Conditioning - Occurs when a behavior is associated with a significant event.



Reinforcement and Punishment

	ADD SOMETHING	TAKE AWAY SOMETHING
INCREASE BEHAVIOR	Positive Reinforcement	Negative Reinforcement
DECREASE BEHAVIOR	Positive Punishment	Negative Punishment

- Reinforcers *increase* behavior
- Punishers *decrease* behavior



Operant Conditioning

- "The Big Bang Theory" OC Example

- **Discussion: How might you use OC in your own life? How has OC been used on you in the past?**

3 Things to Know About Instrumental (Operant) Conditioning



1. Instrumental Responses can occur under stimulus control.

2. Operant Conditioning involves choice.

3. Reinforcers are not made equal.

From Conditioning to Social-Cognitive Learning

What we have covered

Classical conditioning: learning through associations between stimuli.

Operant conditioning: learning through consequences of behavior.

Both approaches explain important forms of adaptive behavior.

Where we go next

Learners do more than respond automatically to stimuli and rewards.

They form mental representations of the environment.

They can learn without immediate reinforcement.

They can learn by observing models and consequences for others.

Learning is not only behavioral; it is also cognitive and social.

Cognitive-Social Approaches to Learning

Tolman's "latent learning"

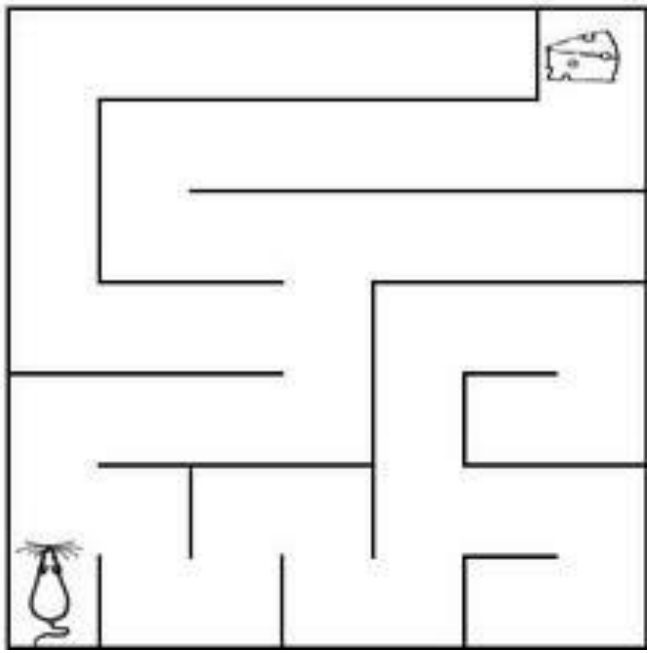
- Tolman: demonstrated "latent learning" using an operant conditioning paradigm
  - Rats explored a maze with no reward
  - Later, under conditions of reward: could demonstrate formation of a "cognitive map" – a mental representation of spatial locations and directions
  - Indicated that learning had taken place, not "mere" conditioning

E.C. Tolman

- Cognitive Theory of Learning
  - Learning as developing from bits of knowledge and cognitions about the environment and how the organism relates to it.
  - This was in contrast to the theories of Thorndike who thought of learning as a strict stimulus-response connection.

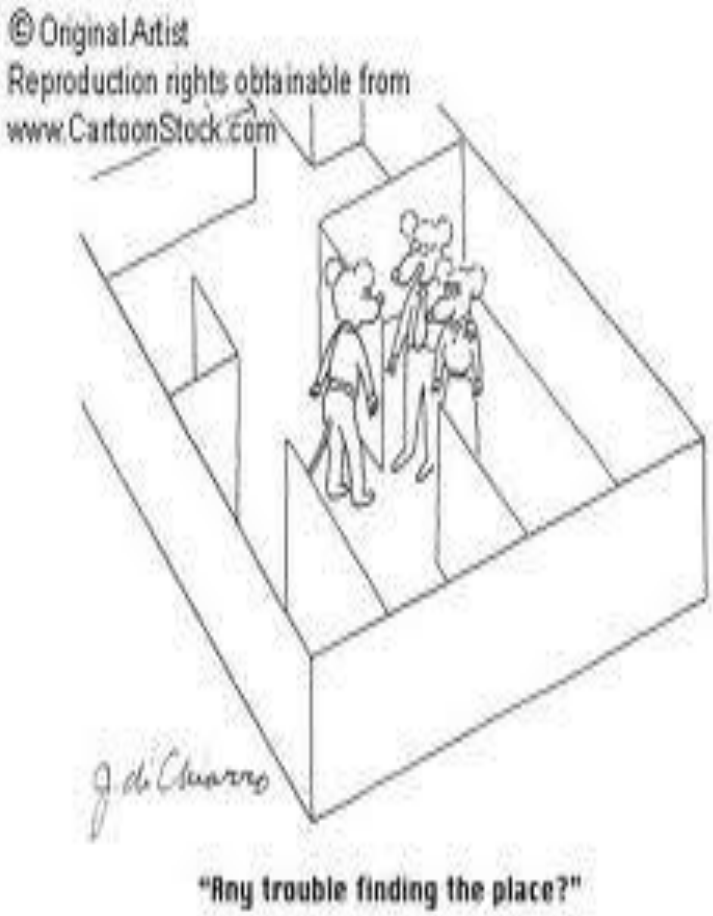
Cognitive Maps

A mental representation of the layout of one's environment.



Rats and Mazes

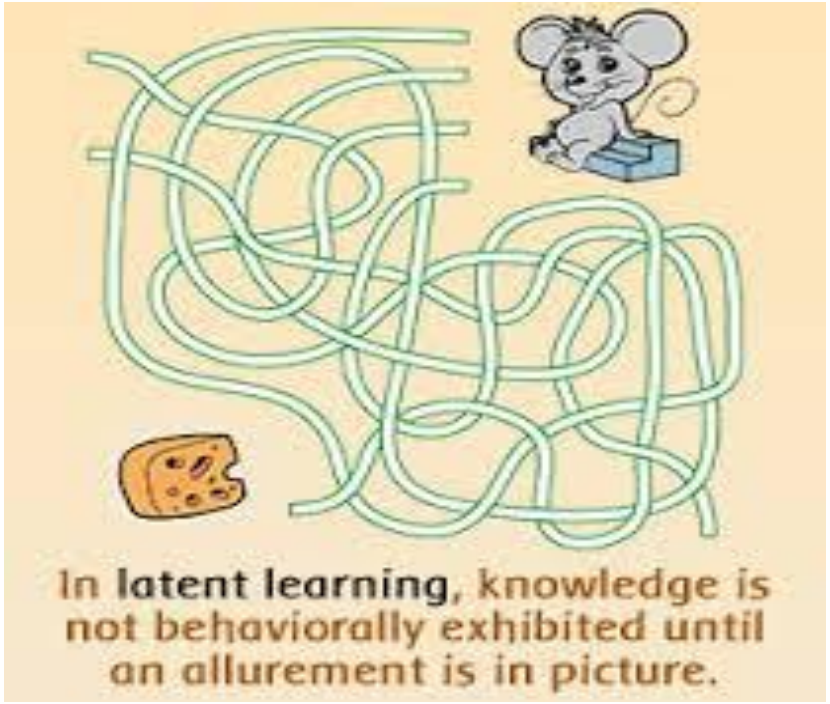
- Tolman conducted experiments with rats and mazes to examine the role that reinforcement plays in the way that rats learn their way through complex mazes. These experiments eventually led to the theory of latent learning



Latent Learning

- Latent learning which describes learning that occurs in the absence of an obvious reward
- Tolman argued that humans engage in this type of learning everyday as we drive or walk the same route daily and learn the locations of various buildings and objects.
- Only when we need to find a building or object does learning become obvious

Tolman's "latent learning"





Is there latent learning in the film "**Kung Fu Panda**"?

Observational Learning – Albert Bandura

- Learning by observing the behavior of another person, or model.
- The Model - the person whose behavior the participant watches and imitates
- Modeling - The process of observing and imitating a specific behavior

Albert Bandura

- An influential American psychologist who is known for his work in observational learning, self-efficacy, moral dis-engagement.
- Studied the consequences a model has on participants
- Bobo Doll experiments
- Read - https://en.wikipedia.org/wiki/Albert_Bandura



Bobo Doll Experiments

- Children watched an adult model show aggressive behavior toward a Bobo doll

- Three experimental conditions:
  - The model was praised.
  - The model was punished.
  - The model received no consequences for the aggressive behavior.

Vicarious Learning

- Learning by seeing the consequence of another's behavior

- Bandura said that reinforcement is not essential for learning to occur, but the expectation of reinforcement affects the performance of what has been learned.

Bobo Doll Experiment














Bandura’s Bobo Dolls



Violence in Television and Video games: Does the media's message matter?

Violence in Television and Video games: Does the media's message matter?

- People frequently learn and imitate the aggression they observe.
- Playing violent video games found to be linked with actual aggression.
- Exposure to media violence may contribute to real life aggression
- Supported by significant psychological research (See Feldman 2013, p. 202)

What Have We Learned About Learning?

- **Classical conditioning:** organisms learn that one stimulus predicts another.
- **Operant conditioning:** organisms learn which behaviors bring reward or punishment.
- **Social-cognitive approaches:** organisms learn by forming cognitive maps, observing models, and anticipating outcomes.
- Together, these approaches show that learning is **associative, behavioral, cognitive, and social**.